

# Probability Toolkit

Probability is a fraction of the whole: everything adds to 1. Expected number = probability  $\times$  trials. Count first, then divide.



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## THE TOPIC ON ONE MAP

### Probability Toolkit

#### 01 Probability tables

add to 1   find x    $P(\text{not } A)$

#### 02 Expected and relative frequency

$P \times n$     $\frac{f}{n}$    more trials

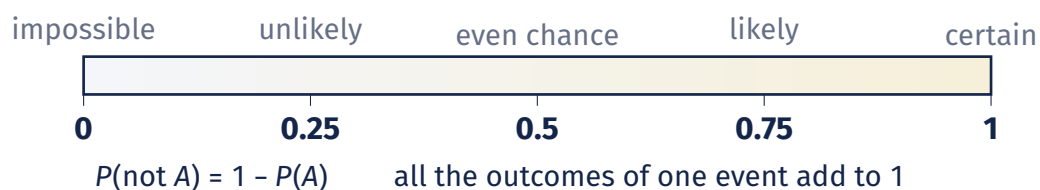
#### 03 How many in the bag?

total =  $\frac{\text{count}}{P}$    ratios   add counters

#### 04 Sample spaces and two picks

outcome table   with / without replacement

**THE SCALE** Every probability sits between 0 and 1:



# 01 Probability Tables: Everything Adds to 1

A table lists every outcome of one event, so the probabilities must add up to exactly 1. The missing number is  $1 -$  (the ones you know). If there are two missing, use the extra fact to write an equation.

## WHEN YOU SEE → YOU DO

a table with one gap

→ gap =  $1 -$  sum of the others

“the probability of red is twice blue”

→ red =  $2x$ , blue =  $x$ , then add to 1

“probability of A or B”

→ add them (one pick cannot be both)

“probability of not A”

→  $1 - P(A)$

a biased dice with  $2x$ ,  $0.18$ ,  $3x$ , ...

→ collect the  $x$  terms, solve

## WORKED MODEL • A BIASED DICE

Probabilities:  $2x$ ,  $0.18$ ,  $2x$ ,  $3x$ ,  $0.26$ ,  $x$  for 1 to 6. Find  $x$  and  $P(\text{even})$ .

1  $8x + 0.44 = 1 \Rightarrow x = 0.07$

add to 1

2 even:  $0.18 + 3x + x = 0.18 + 0.28$

numbers 2, 4, 6

ANSWER  $P(\text{even}) = 0.46$

× **TRAP** Adding to 100 instead of 1. Decimals add to 1; percentages add to 100%.

IDEA 01

## Practise: Probability Tables: Everything Adds to 1

My questions first, then selected practice, then real past papers.

**Q1** **DR ESLAM** Missing probability

CH 6 • TOPIC 4 2 marks

Probabilities: red 0.3, blue 0.25, green  $x$ , yellow  $2x$ . Find  $x$ .

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## SELECTED PRACTICE

**Q2****PRACTICE**

Practice • Easy

CH 6 • TOPIC 4 4 marks

- 3 (a)** There are some green counters, some yellow counters, some blue counters and some red counters in a bag.

The table shows the probabilities that a counter taken at random from the bag will be green or yellow or red.

| Colour      | Green | Yellow | Blue | Red  |
|-------------|-------|--------|------|------|
| Probability | 0.16  | 0.4    |      | 0.24 |

Mary takes at random a counter from the bag.

Work out the probability that the counter will be blue.

**(2 marks)**

- (b)** Mary puts the counter back into the bag.  
There are 125 counters in the bag.

Work out the number of green counters in the bag.

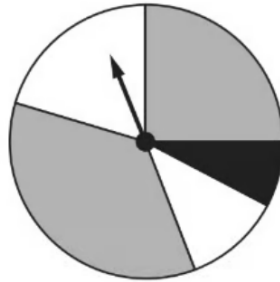
**(2 marks)**

**Q3****PRACTICE**

Practice • Easy

**CH 6 • TOPIC 4** 1 mark

- 13** This spinner has two grey sections, two white sections and one black section.



Vlad says

The probability of the spinner landing on black is  $\frac{1}{5}$ .

Explain why Vlad is not correct.

(1 mark)

**Q4****PRACTICE**

Practice • Medium

CH 6 • TOPIC 4 6 marks

**14 (a)** In a bag there are only red bricks, blue bricks, green bricks and orange bricks.

The number of green bricks in the bag is the same as the number of orange bricks.

Jiao takes at random a brick from the bag.

The table gives the probability that Jiao takes a red brick and the probability that he takes a blue brick.

| Colour      | red  | blue | green | orange |
|-------------|------|------|-------|--------|
| Probability | 0.26 | 0.3  |       |        |

Work out the probability that Jiao takes an orange brick.

**(3 marks)**

**(b)** Jiao puts the brick back into the bag.

There are 91 red bricks in the bag.

Jiao is going to build a tower using all the red bricks and all the blue bricks but no other bricks.

The tower will be in the shape of a cuboid.

There will be 4 bricks in each layer of the tower.

Work out how many layers the tower will have.

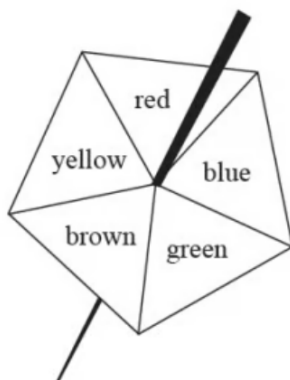
**(3 marks)**

**Q5****PRACTICE**

Practice • Hard

CH 6 • TOPIC 4 4 marks

16 Here is a biased 5-sided spinner.



When the spinner is spun, it can land on red, blue, green, brown or yellow.

The table gives the probabilities that the spinner lands on red or on blue or on green.

| Colour      | red  | blue | green | brown | yellow |
|-------------|------|------|-------|-------|--------|
| Probability | 0.15 | 0.26 | 0.33  |       |        |

When the spinner is spun once, the probability that the spinner lands on brown is **0.06** more than the spinner lands on yellow.

Jennie spins the spinner **150** times.

Work out an estimate for the number of times the spinner lands on yellow.

**(4 marks)**

## PAST PAPERS

**Q6****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q6

**CH 6 • TOPIC 4 4 marks**

- 6 A bag contains only red beads, blue beads, green beads and yellow beads.

The table gives the probabilities that, when a bead is taken at random from the bag, the bead will be blue or the bead will be yellow.

| Colour      | red | blue | green | yellow |
|-------------|-----|------|-------|--------|
| Probability |     | 0.24 |       | 0.31   |

The probability that the bead will be green is twice the probability that the bead will be red.

Sofia takes at random a bead from the bag.

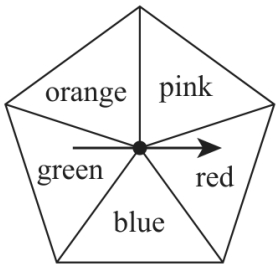
She writes down the colour of the bead and puts the bead back into the bag.

She does this 180 times.

Work out an estimate for the number of times she takes a red bead from the bag.

(Total for Question 6 is 4 marks)

6 Grace has a biased 5-sided spinner.



Grace is going to spin the arrow on the spinner once.

The table below gives the probabilities that the spinner will land on red or on blue or on green.

| Colour      | Red  | Blue | Green | Orange | Pink |
|-------------|------|------|-------|--------|------|
| Probability | 0.20 | 0.12 | 0.08  |        |      |

The probability that the spinner will land on orange is 3 times the probability that the spinner will land on pink.

(a) Work out the probability that the spinner will land on orange.

.....

Grace spins the arrow on the spinner 150 times.

(b) Work out an estimate for the number of times the spinner lands on blue.

.....

(Total for Question 6 is 5 marks)

**Q8****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q1

**CH 6 • TOPIC 4 4 marks****You must write down all the stages in your working.**

- 1 A tin contains tea bags with a choice of four different flavours of tea.  
The four flavours of tea are Assam or Darjeeling or Nilgiri or Rize.

Sara takes at random a tea bag from the tin.

The table shows each of the probabilities that the flavour of the tea Sara takes is Assam or Darjeeling or Rize.

| Flavour of tea | Assam | Darjeeling | Nilgiri | Rize |
|----------------|-------|------------|---------|------|
| Probability    | 0.38  | 0.24       |         | 0.16 |

- (a) Work out the probability that the flavour of the tea Sara takes is Nilgiri.

.....

- (b) Work out the probability that the flavour of the tea Sara takes is either Darjeeling or Rize.

.....

**(Total for Question 1 is 4 marks)**

**Q9****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q6

**CH 6 • TOPIC 4 5 marks**

- 6 A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

| Sweet       | pink | white | green | red  |
|-------------|------|-------|-------|------|
| Probability |      |       | 0.2   | 0.35 |

The ratio

number of pink sweets : number of white sweets = 2 : 1

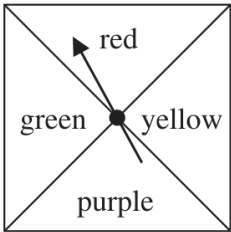
There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

(Total for Question 6 is 5 marks)

You must write down all the stages in your working.

1 Here is a biased spinner.



When the spinner is spun once, the probabilities that it lands on red or on yellow or on green are given in the table.

|             |      |        |        |       |
|-------------|------|--------|--------|-------|
| Colour      | red  | yellow | purple | green |
| Probability | 0.25 | 0.2    |        | 0.2   |

(a) Work out the probability that the spinner lands on red or on yellow.

Yang is going to spin the spinner 300 times.

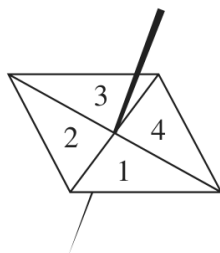
(b) Work out an estimate for the number of times the spinner will land on purple.

(Total for Question 1 is 4 marks)

**Q11** PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1H • Q2

CH 6 • TOPIC 4 4 marks

2 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

| Number      | 1    | 2 | 3    | 4 |
|-------------|------|---|------|---|
| Probability | 0.26 |   | 0.18 |   |

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

(Total for Question 2 is 4 marks)

**Q12****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q2

**CH 6 • TOPIC 4 3 marks**

- 2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

| Colour      | green | yellow | brown | pink |
|-------------|-------|--------|-------|------|
| Probability | 0.32  | 0.13   | 0.28  |      |

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

.....  
(Total for Question 2 is 3 marks)

**Q13****PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q4

**CH 6 • TOPIC 4 4 marks**

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

|                    |      |        |       |
|--------------------|------|--------|-------|
| <b>Colour</b>      | blue | orange | white |
| <b>Probability</b> | 0.58 | $2x$   | $x$   |

- (a) Work out the value of  $x$

$x =$  .....

The spinner is spun 250 times.

- (b) Work out an estimate for the number of times the spinner will land on blue.

.....

**(Total for Question 4 is 4 marks)**

Q14

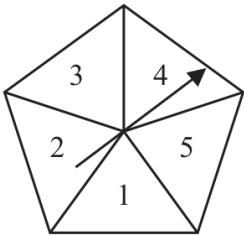
PAST PAPER

Edexcel 4MA1 • May 2024 • Paper 1HR • Q2

CH 6 • TOPIC 44 marks

PRACTISE • 6.4

2 Here is a biased spinner.



The table gives information about the probability that, when the spinner is spun once, it will land on each number.

|             |      |      |      |     |      |
|-------------|------|------|------|-----|------|
| Number      | 1    | 2    | 3    | 4   | 5    |
| Probability | $2x$ | 0.27 | 0.04 | $x$ | 0.12 |

Alexis is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on an odd number.

.....

(Total for Question 2 is 4 marks)

.....

**Q15****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q2

**CH 6 • TOPIC 4 4 marks**

- 2 450 students were asked how they travelled to school on Monday.  
Each student walked or travelled by bus or travelled by car or travelled by bicycle.  
Each student used just one method of travel.

One of these students is chosen at random.

The table shows information about the probability of each method of travel.

| Method of travel | walk | bus | car  | bicycle |
|------------------|------|-----|------|---------|
| Probability      | 0.20 | $x$ | $2x$ | 0.26    |

Work out how many of the 450 students travelled by car.

(Total for Question 2 is 4 marks)

Q16

PAST PAPER

Edexcel 4MA1 · November 2024 · Paper 2H · Q5

CH 6 · TOPIC 4

4 marks

PRACTISE · 6.4

5 Cody has a biased 5-sided spinner, numbered 1, 2, 3, 4, 5

The table gives the probabilities that when the spinner is spun it will land on 2 or on 3 or on 5

|             |   |      |      |   |      |
|-------------|---|------|------|---|------|
| Number      | 1 | 2    | 3    | 4 | 5    |
| Probability |   | 0.14 | 0.17 |   | 0.21 |

The probability that the spinner will land on 1 is the same as the probability that the spinner will land on 4

Cody is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on 4

(Total for Question 5 is 4 marks)

Q17

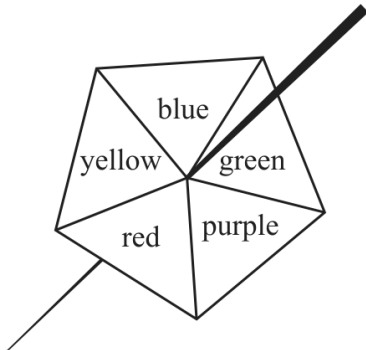
PAST PAPER

Edexcel 4MA1 • June 2025 • Paper 1HR • Q4

CH 6 • TOPIC 45 marks

PRACTISE • 6.4

- 4 Here is a biased 5-sided spinner.  
When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.



The table gives information about the probability of the spinner landing on each colour.

|             |      |       |        |      |        |
|-------------|------|-------|--------|------|--------|
| Colour      | blue | green | purple | red  | yellow |
| Probability | 0.12 | 0.20  | 0.38   | $4x$ | $x$    |

Sophie spins the spinner once.

- (a) Work out the probability that the spinner lands on blue or on green or on purple.

.....

Max spins the spinner 350 times.

- (b) Work out an estimate for the number of times the spinner lands on red.

.....

(Total for Question 4 is 5 marks)

## 02 Expected and Relative Frequency

Expected number = probability  $\times$  number of trials. Relative frequency =  $\frac{\text{times it happened}}{\text{number of trials}}$  — it is an **estimate** of the probability, and more trials give a better estimate.

### WHEN YOU SEE $\rightarrow$ YOU DO

“estimate the number of times”

$\rightarrow P \times n$

“the coin is thrown 500 times, expected tails”

$\rightarrow P(\text{tails}) = 1 - P(\text{heads})$  first

results of an experiment

$\rightarrow$  relative frequency =  $\frac{f}{n}$

“whose results are most reliable?”

$\rightarrow$  the person with the most trials

a game: cost to play and a prize

$\rightarrow$  money in – expected prizes paid out

Find the probability

Times the trials

Round sensibly

### WORKED MODEL • A CHARITY GAME

$P(\text{win}) = \frac{1}{4}$ . It costs 30p to play, the prize is £1. Played 100 times. Money raised?

1 in:  $100 \times 30\text{p} = \text{£}30$

money taken

2 expected wins:  $\frac{1}{4} \times 100 = 25$ , paid out £25

$P \times n$

ANSWER  $\text{£}30 - \text{£}25 = \text{£}5$

## IDEA 02

**Practise: Expected and Relative Frequency**

My questions first, then selected practice, then real past papers.

**Q18** **DR ESLAM** Expected

CH 6 • TOPIC 4 2 marks

$P(\text{win}) = 0.35$ . Estimate the number of wins in 240 games.

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**SELECTED PRACTICE****Q19** **PRACTICE** Practice • Easy

CH 6 • TOPIC 4 2 marks

- 1 The probability that a biased dice will land on a five is 0.3

Megan is going to roll the dice 400 times.

Work out an estimate for the number of times the dice will land on a five.

**(2 marks)**

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**Q20****PRACTICE**

Practice · Easy

**CH 6 · TOPIC 4 1 mark**

- 11** The probability of Heads when a biased coin is thrown is 0.6

The coin is thrown 500 times.

Circle the expected number of Tails.

|    |     |     |     |
|----|-----|-----|-----|
| 20 | 200 | 250 | 300 |
|----|-----|-----|-----|

(1 mark)

**Q21****PRACTICE**

Practice • Medium

**CH 6 • TOPIC 4 3 marks**

- 6 The table shows the probabilities that a biased dice will land on 2, on 3, on 4, on 5 and on 6

| Number on dice | 1 | 2    | 3    | 4    | 5    | 6   |
|----------------|---|------|------|------|------|-----|
| Probability    |   | 0.17 | 0.18 | 0.09 | 0.15 | 0.1 |

Neymar rolls the biased dice 200 times.

Work out an estimate for the total number of times the dice will land on 1 or on 3.

**(3 marks)**

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**Q22****PRACTICE**

Practice • Hard

**CH 6 • TOPIC 4 5 marks**

- 9 (a)** Four friends each throw a biased coin a number of times. The table shows the number of heads and the number of tails each friend got.

|       | Ben | Helen | Paul | Sharif |
|-------|-----|-------|------|--------|
| heads | 34  | 66    | 80   | 120    |
| tails | 8   | 12    | 40   | 40     |

The coin is to be thrown one more time.

Which of the four friends' results will give the best estimate for the probability that the coin will land heads?

Justify your answer.

**(1 mark)**

- (b)** Paul says,

"With this coin you are twice as likely to get heads as to get tails."

Is Paul correct?

Justify your answer.

**(2 marks)**

- (c)** The coin is to be thrown twice.

Use all the results in the table to work out an estimate for the probability that the coin will land heads both times.

**(2 marks)**

## PAST PAPERS

**Q23****PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q2

**CH 6 • TOPIC 4 4 marks**

- 2 Toy cars are made in a factory.  
The toy cars are made for 15 hours each day.  
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

(Total for Question 2 is 4 marks)

**Q24****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q2

**CH 6 • TOPIC 4 2 marks**

- 2 Each time John plays a game, the probability that he wins the game is 0.65

John is going to play the game 300 times.

Work out an estimate for the number of games that John wins.

(Total for Question 2 is 2 marks)

**Q25****PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q5

**CH 6 • TOPIC 4 2 marks**

- 5 Each time Evie plays a game against her computer, she will win or lose.

For each game, the probability that Evie will win is 0.74  
Evie is going to play 300 games against her computer.

Work out an estimate for the number of games that Evie will lose.

(Total for Question 5 is 2 marks)

## 03 How Many in the Bag?

A probability is  $\frac{\text{how many of that kind}}{\text{how many altogether}}$ . So if you know the count and the probability, the total is  $\text{count} \div P$ . When counters are added, the total changes too.

### WHEN YOU SEE → YOU DO

“12 red counters,  $P(\text{red}) = \frac{2}{5}$ , how many altogether?”

$$\rightarrow 12 \div \frac{2}{5} = 30$$

“the ratio red : blue is 3 : 17”

→ split the part that is not purple in that ratio

“some more white fish are added, now  $P = \frac{1}{2}$ ”

$$\rightarrow \frac{w+k}{n+k} = \frac{1}{2} \text{ — both change}$$

“twice as many blue as yellow”

→ yellow =  $y$ , blue =  $2y$  — write everything in  $y$

### WORKED MODEL • NOVEMBER 2021 P1H

54 fish,  $P(\text{white}) = \frac{4}{9}$ . Then  $k$  white fish are added and  $P(\text{white}) = \frac{1}{2}$ . Find  $k$ .

1 white =  $\frac{4}{9} \times 54 = 24$

count =  $P \times$  total

2  $\frac{24+k}{54+k} = \frac{1}{2} \Rightarrow 48 + 2k = 54 + k$

both numbers change

**ANSWER**  $k = 6$  white fish

× **TRAP** Keeping the old total after adding counters. The bottom of the fraction grows too.

## IDEA 03

**Practise: How Many in the Bag?**

My questions first, then selected practice, then real past papers.

**Q26** **DR ESLAM** In the bag

CH 6 • TOPIC 4 3 marks

A bag has 12 red counters and some blue.  $P(\text{red}) = \frac{2}{5}$ . How many blue counters?

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**SELECTED PRACTICE****Q27** **PRACTICE** Practice • Medium

CH 6 • TOPIC 4 3 marks

- 1 One of the teachers at a school is chosen at random.

The probability that this teacher is female is  $\frac{3}{5}$

There are 36 **male** teachers at the school.

Work out the total number of teachers at the school.

(3 marks)

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**Q28****PRACTICE**

Practice • Hard

**CH 6 • TOPIC 4 3 marks**

- 5 There are only blue cubes, yellow cubes and green cubes in a bag.

There are

|     |                                               |
|-----|-----------------------------------------------|
|     | twice as many blue cubes as yellow cubes      |
| and | four times as many green cubes as blue cubes. |

Hannah takes at random a cube from the bag.

Work out the probability that Hannah takes a yellow cube.

**(3 marks)**

**Q29****PRACTICE**

Practice • Hard

**CH 6 • TOPIC 4** 3 marks

- 8 There are only red counters, blue counters and purple counters in a bag.  
The ratio of the number of red counters to the number of blue counters is 3 : 17  
Sam takes at random a counter from the bag.  
The probability that the counter is purple is 0.2

Work out the probability that Sam takes a red counter.

**(3 marks)**

**Q30****PRACTICE**

Practice • Hard

CH 6 • TOPIC 4

**4 marks**

**18** A bag contains some counters.

- There are 300 counters in the bag.
- There are only red, white and blue counters in the bag.
- The probability of picking a blue counter is  $\frac{23}{50}$
- The ratio of red counters to white counters is 2 : 1.

Calculate the number of red counters in the bag.

**(4 marks)**

## PAST PAPERS

**Q31** PAST PAPER Edexcel 4MA1 • November 2021 • Paper 1H • Q3

CH 6 • TOPIC 4 4 marks

- 3 There are 54 fish in a tank.  
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is  $\frac{4}{9}$

- (a) Work out the number of white fish originally in the tank.

.....

Jeevan puts the fish he took out, back into the tank.

He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is now  $\frac{1}{2}$

- (b) Work out the number of white fish Jeevan put into the tank.

.....

(Total for Question 3 is 4 marks)

**Q32****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q11

**CH 6 • TOPIC 4 3 marks**

11 Khalid has a box of counters.

17 of the counters are red

28 of the counters are blue

the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is  $\frac{4}{9}$

Work out the number of orange counters that are in the box.

.....  
(Total for Question 11 is 3 marks)

**Q33****PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q9

**CH 6 • TOPIC 4 3 marks**

- 9 Mark has three bags, bag A, bag B and bag C

Bag A contains 120 counters.

The probability of taking at random a red counter from bag A is 0.45

Bag B contains 80 counters.

The probability of taking at random a red counter from bag B is 0.3

Bag C is empty.

Mark puts all the counters from bag A and bag B into bag C

Mark takes at random a counter from bag C

Work out the probability that he takes a red counter.

.....  
(Total for Question 9 is 3 marks)

## 04 Sample Spaces and Two Picks

For two things happening (two dice, two spinners, two cards), draw a sample space table: every box is one equally likely outcome. Count the boxes you want and divide by the total.

### WHEN YOU SEE → YOU DO

**two dice, “score = sum / difference / product”**

→ table, 36 boxes

**“list all the possible outcomes”**

→ systematic list — one row at a time

**“takes a card, does not replace it”**

→ second pick: one fewer on top and bottom

**“probability both are ...”**

→ multiply the two probabilities

**“at least one”**

→  $1 - P(\text{none})$

| + | 1 | 2 | 3 | 4  | 5  | 6  |
|---|---|---|---|----|----|----|
| 1 | 2 | 3 | 4 | 5  | 6  | 7  |
| 2 | 3 | 4 | 5 | 6  | 7  | 8  |
| 3 | 4 | 5 | 6 | 7  | 8  | 9  |
| 4 | 5 | 6 | 7 | 8  | 9  | 10 |
| 5 | 6 | 7 | 8 | 9  | 10 | 11 |
| 6 | 7 | 8 | 9 | 10 | 11 | 12 |

$$P(\text{total } 7) = \frac{6}{36} = \frac{1}{6}$$

### WORKED MODEL • WITHOUT REPLACEMENT

5 red and 3 blue counters. Two are taken without replacement.  $P(\text{both red})$ ?

1 first red:  $\frac{5}{8}$ ; second red:  $\frac{4}{7}$

one red and one counter gone

**ANSWER**  $\frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$

## IDEA 04

**Practise: Sample Spaces and More Than One Pick**

My questions first, then selected practice, then real past papers.

**Q34** DR ESLAM Without replacement

CH 6 • TOPIC 4 3 marks

5 red and 3 blue counters. Two are taken without replacement. Find  $P(\text{both red})$ .

---

SELECTED PRACTICE

**Q35**

PRACTICE Practice • Easy

CH 6 • TOPIC 4 2 marks

PRACTISE • 6.4

12 (a) Dora has the following number cards.

2

2

3

5

6

She takes a card at random, replaces the card and then takes a second card.  
She adds the numbers on the two cards she has taken and records the total.  
Complete the following table to show all of her possible totals.

|             |       | First card |   |   |    |    |
|-------------|-------|------------|---|---|----|----|
| Second card | Total | 2          | 2 | 3 | 5  | 6  |
|             | 2     | 4          | 4 | 5 | 7  | 8  |
|             | 2     | 4          | 4 | 5 |    | 8  |
|             | 3     | 5          | 5 |   | 8  | 9  |
|             | 5     | 7          |   | 8 | 10 | 11 |
|             | 6     | 8          | 8 | 9 | 11 | 12 |

(1 mark)

(b) Find the probability that her total is

i) an even number,

[2]

ii) a multiple of 3 or 4.

[2]

(1 mark)

Q36

PRACTICE

Practice • Easy

CH 6 • TOPIC 44 marks

PRACTISE • 6.4

16 This is a fair 5-sided spinner.



Ciara spins the spinner twice and records the product of the two scores.

i) Complete the table.

| Second spin | First spin |   |   |   |    |   |
|-------------|------------|---|---|---|----|---|
|             | x          | 1 | 2 | 2 | 3  | 4 |
|             | 1          | 1 |   |   |    |   |
|             | 2          |   |   | 4 |    |   |
|             | 2          |   |   |   |    |   |
|             | 3          |   |   |   |    |   |
|             | 4          |   |   |   | 12 |   |

[2]

ii) Find the probability that the product is a multiple of 3.

[2]

(4 marks)

**Q37****PRACTICE**

Practice · Medium

CH 6 · TOPIC 4 4 marks

**10 (a)**  $\xi = \{\text{even numbers}\}$  $A = \{\text{factors of 8}\}$  $B = \{\text{factors of 20}\}$ List the members of the set  $A \cup B$ .**(2 marks)****(b)** A number is chosen at random from set B.Find the probability that the number is in set  $(A \cap B)$ **(2 marks)**

Q38

PRACTICE

Practice • Medium

CH 6 • TOPIC 4

2 marks

PRACTISE • 6.4

15 Antonio rolls two fair six-sided dice and calculates the **difference** between the scores. For example, if the two scores are 2 and 5 or 5 and 2 then the difference is 3.

Complete the sample space diagram to show the possible outcomes from Antonio's dice.

Dice 2

| Dice 1 | difference | 1 | 2 | 3 | 4 | 5 | 6 |
|--------|------------|---|---|---|---|---|---|
|        | 1          | 0 |   |   |   |   |   |
|        | 2          |   |   |   |   | 3 |   |
|        | 3          |   | 1 |   |   |   |   |
|        | 4          |   |   |   |   |   |   |
|        | 5          |   | 3 |   |   |   |   |
|        | 6          |   |   |   |   |   |   |
|        |            |   |   |   |   |   |   |

(2 marks)

## PAST PAPERS

**Q39****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q18

**CH 6 • TOPIC 4 • 4 marks**

**18** There are 16 sweets in a bowl.

4 of the sweets are blackcurrant.

5 of the sweets are lemon.

7 of the sweets are orange.

Anna, Ravi and Sam each take at random one sweet from the bowl.

Work out the probability that the 5 lemon sweets are still in the bowl.

.....  
(Total for Question 18 is 4 marks)

**Q40****PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q23

**CH 6 • TOPIC 4 4 marks**

23 In a bag, there are only

3 blue beads  
4 white beads  
and  $x$  orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is  $\frac{3}{8}$

Find the total number of beads in the bag.  
Show clear algebraic working.

.....  
(Total for Question 23 is 4 marks)

**Q41****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q21

**CH 6 • TOPIC 4 6 marks**

- 21 A bag contains  $n$  beads.  
6 of the beads are red and the rest are blue.

Ravi is going to take at random 2 beads from the bag.

The probability that the 2 beads will be of the same colour is  $\frac{9}{17}$

Using algebra, and showing each stage of your working, calculate the value of  $n$ .

$n = \dots\dots\dots$

**(Total for Question 21 is 6 marks)**

**Q42** **PAST PAPER** Edexcel 4MA1 • May 2021 • Paper 2H • Q23**CH 6 • TOPIC 4** **5 marks**

23 Pippa has a box containing  $N$  pens.

There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is  $\frac{9}{35}$

Find the possible values of  $N$ .

Show clear algebraic working.

(Total for Question 23 is 5 marks)

**Q43****PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q24

**CH 6 • TOPIC 4 5 marks**

**24** Elliot has  $x$  counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.  
He then picks at random a second counter and turns the counter over.

The probability that there are still 5 counters showing a red face is  $\frac{19}{32}$

Work out the value of  $x$   
Show clear algebraic working.

$x =$  .....

**(Total for Question 24 is 5 marks)**

**Q44** **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 1H • Q15**CH 6 • TOPIC 4** **4 marks**

- 15** Abraham is going to play a computer game.  
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3  
the probability that he draws the game is 0.5  
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.

When Abraham draws a game, he scores 0 points.

When Abraham loses a game, he scores −5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

.....

**(Total for Question 15 is 4 marks)**

**Q45****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q15

**CH 6 • TOPIC 4 3 marks**

- 15 Osvaldo has a biased coin.  
He spins the coin three times.

The probability that the coin lands on a head three times is  $\frac{27}{64}$

Work out the probability that the coin will land on a tail three times.

(Total for Question 15 is 3 marks)

**Q46****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q19

**CH 6 • TOPIC 4 4 marks**

19 30 adults booked to stay in a hotel.

19 adults booked breakfast

15 adults booked dinner

4 adults did not book breakfast or dinner

Some adults booked breakfast **and** dinner.

Meihui chooses at random two of the 30 adults.

Work out the probability that these two adults each booked breakfast **and** dinner.

(Total for Question 19 is 4 marks)

**Q47****PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q20

**CH 6 • TOPIC 4 4 marks**

**20** A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.  
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is  $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

.....

**(Total for Question 20 is 4 marks)**

**Q48****PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q14

**CH 6 • TOPIC 4 3 marks**

**14** Adriana is going to roll a biased dice and spin a biased coin.

The probability that the coin will land on Heads is 0.8

The probability that the dice will land on 6 and the coin will land on Heads is 0.24

Work out the probability that the dice will land on 6 and the coin will land on Tails.

.....

**(Total for Question 14 is 3 marks)**

**Q49****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q21

**CH 6 • TOPIC 4 5 marks**

21 There are 25 counters in a bag such that

6 counters are blue

$x$  counters are orange, where  $x > 9$

the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is  $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.

Show clear algebraic working.

.....

**(Total for Question 21 is 5 marks)**

**Q50****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q19

**CH 6 • TOPIC 4 3 marks****19** Kannika has 9 counters.

There is a number on each counter.



Kannika puts the 9 counters in a bag.

She takes at random a counter from the bag and does not replace the counter.

She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters is less than 5

.....

(Total for Question 19 is 3 marks)

**Q51****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q21

**CH 6 • TOPIC 4 3 marks**

**21** A box contains 20 counters.

- 9 of the counters are red
- 7 of the counters are yellow
- 4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....

**(Total for Question 21 is 3 marks)**

# ALL The Whole Topic on One Page

Tear this page out before the exam.

- |           |                   |                                                                         |
|-----------|-------------------|-------------------------------------------------------------------------|
| <b>01</b> | <b>Tables</b>     | Everything adds to 1; the gap is 1 – the rest.                          |
| <b>02</b> | <b>Expected</b>   | $P \times n$ ; relative frequency $\frac{f}{n}$ ; more trials = better. |
| <b>03</b> | <b>In the bag</b> | Total = count $\div P$ ; adding changes top and bottom.                 |
| <b>04</b> | <b>Two picks</b>  | Sample space table; without replacement $\rightarrow$ one fewer.        |

**BOTTOM LINE** Add to 1 • expected =  $P \times n$  • count  $\div$  total • two picks  $\rightarrow$  table or multiply.

# Answers

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Probability Toolkit • for the teacher. Every answer checked with a computer algebra system.

## IDEA 01 • PROBABILITY TABLES: EVERYTHING ADDS TO 1

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|            |                                                                                                     |                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Q1</b>  | Dr Eslam • Missing probability                                                                      | $3x = 0.45, x = 0.15$                                                                                       |
| <b>Q2</b>  | Practice • Easy                                                                                     | (a) $1 - 0.8 = 0.2$ (b) $0.16 \times 125 = 20$                                                              |
| <b>Q3</b>  | Practice • Easy                                                                                     | The sections are not the same size, so the outcomes are not equally likely                                  |
| <b>Q4</b>  | Practice • Medium                                                                                   | (a) $(1 - 0.56) \div 2 = 0.22$ (b) total = $91 \div 0.26 = 350$ ; blue 105; $(91 + 105) \div 4 = 49$ layers |
| <b>Q5</b>  | Practice • Hard                                                                                     | brown + yellow = 0.26, brown = yellow + 0.06: yellow = 0.10; $150 \times 0.1 = 15$                          |
| <b>Q6</b>  | Edexcel 4MA1 • January 2020 • Paper 2H • Q6                                                         | 27                                                                                                          |
| <b>Q7</b>  | Edexcel 4MA1 • May 2021 • Paper 1H • Q6                                                             | Pink 0.15, orange 0.45, blue 18                                                                             |
| <b>Q8</b>  | Edexcel 4MA1 • January 2022 • Paper 2HR • Q1                                                        | Nilgiri 0.22, Darjeeling or Rize 0.40                                                                       |
| <b>Q9</b>  | Edexcel 4MA1 • January 2022 • Paper 2H • Q6                                                         | 12                                                                                                          |
| <b>Q10</b> | Edexcel 4MA1 • June 2022 • Paper 1HR • Q1                                                           | 0.45, 0.35, 105                                                                                             |
| <b>Q11</b> | Edexcel 4MA1 • June 2022 • Paper 1H • Q2                                                            | 0.28, 250 spins, 70                                                                                         |
| <b>Q12</b> | Edexcel 4MA1 • June 2023 • Paper 2H • Q2                                                            | 54                                                                                                          |
| <b>Q13</b> | Edexcel 4MA1 • November 2023 • Paper 1H • Q4                                                        | 0.207, and 95                                                                                               |
| <b>Q14</b> | Edexcel 4MA1 • May 2024 • Paper 1HR • Q2                                                            | 0.19, 0.54, 216                                                                                             |
| <b>Q15</b> | Edexcel 4MA1 • May 2024 • Paper 1H • Q2                                                             | 162                                                                                                         |
| <b>Q16</b> | Edexcel 4MA1 • November 2024 • Paper 2H • Q5                                                        | 84                                                                                                          |
| <b>Q17</b> | Dr Eslam Ahmed • eliteigcse.com • WhatsApp 01122900902<br>Edexcel 4MA1 • June 2025 • Paper 1HR • Q4 | 0.62, and about 106                                                                                         |

## IDEA 02 • EXPECTED AND RELATIVE FREQUENCY

- Q18** Dr Eslam • Expected  $0.35 \times 240 = 84$
- Q19** Practice • Easy  $0.3 \times 400 = 120$
- Q20** Practice • Easy  $0.4 \times 500 = 200$
- Q21** Practice • Medium  $P(1) = 0.31; (0.31 + 0.18) \times 200 = 98$
- Q22** Practice • Hard (a) Sharif: the most throws (b) all results: 300 heads, 100 tails — three times as likely, so Paul is not correct (c)  $0.75^2 = \frac{9}{16}$
- Q23** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q2 45
- Q24** Edexcel 4MA1 • January 2022 • Paper 1HR • Q2 195
- Q25** Edexcel 4MA1 • May 2023 • Paper 1HR • Q5 78

## IDEA 03 • HOW MANY IN THE BAG?

- Q26** Dr Eslam • In the bag  
total =  $12 \div \frac{2}{5} = 30$ ; blue = 18
- Q27** Practice • Medium  
 $\frac{2}{5}$  of the teachers = 36: total 90
- Q28** Practice • Hard  
yellow  $y$ , blue  $2y$ , green  $8y$ :  $\frac{1}{11}$
- Q29** Practice • Hard  
red + blue = 0.8:  $\frac{3}{20} \times 0.8 = 0.12$
- Q30** Practice • Hard  
blue = 138; red + white = 162; red = 108
- Q31** Edexcel 4MA1 • November 2021  
• Paper 1H • Q3  
(a) 24 white fish (b)  $\frac{24+k}{54+k} = \frac{1}{2}$ :  $k = 6$  white fish
- Q32** Edexcel 4MA1 • June 2025 • Paper 2H • Q11  
36
- Q33** Edexcel 4MA1 • November 2025  
• Paper 2H • Q9  
0.39

#### IDEA 04 • SAMPLE SPACES AND MORE THAN ONE PICK

|            |                                               |                                                                                 |
|------------|-----------------------------------------------|---------------------------------------------------------------------------------|
| <b>Q34</b> | Dr Eslam • Without replacement                | $\frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$                 |
| <b>Q35</b> | Practice • Easy                               | (a) missing 7, 6, 7 (b)(i) $\frac{13}{25}$ (ii) $\frac{14}{25}$                 |
| <b>Q36</b> | Practice • Easy                               | (i) products of 1, 2, 2, 3, 4 with 1, 2, 2, 3, 4 (ii) $\frac{9}{25}$            |
| <b>Q37</b> | Practice • Medium                             | (a) {2, 4, 8, 10, 20} (b) $A \cap B = \{2, 4\}$ : $\frac{2}{4} = \frac{1}{2}$   |
| <b>Q38</b> | Practice • Medium                             | Row $n$ : $ n - 1 $ , $ n - 2 $ , ..., $ n - 6 $ (e.g. row 4: 3, 2, 1, 0, 1, 2) |
| <b>Q39</b> | Edexcel 4MA1 • January 2020 • Paper 1H • Q18  | $\frac{33}{112}$                                                                |
| <b>Q40</b> | Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q23 | 16 beads                                                                        |
| <b>Q41</b> | Edexcel 4MA1 • January 2021 • Paper 2HR • Q21 | 18                                                                              |
| <b>Q42</b> | Edexcel 4MA1 • May 2021 • Paper 2H • Q23      | $N = 15$ or $N = 21$                                                            |
| <b>Q43</b> | Edexcel 4MA1 • June 2022 • Paper 1HR • Q24    | $x = 8$                                                                         |
| <b>Q44</b> | Edexcel 4MA1 • June 2022 • Paper 1H • Q15     | 0.161                                                                           |
| <b>Q45</b> | Edexcel 4MA1 • June 2023 • Paper 2HR • Q15    | $\frac{1}{64}$                                                                  |
| <b>Q46</b> | Edexcel 4MA1 • June 2023 • Paper 2HR • Q19    | $\frac{28}{435}$                                                                |
| <b>Q47</b> | Edexcel 4MA1 • November 2023 • Paper 1H • Q20 | $\frac{56}{121}$                                                                |
| <b>Q48</b> | Edexcel 4MA1 • June 2024 • Paper 2HR • Q14    | 0.06                                                                            |
| <b>Q49</b> | Edexcel 4MA1 • May 2024 • Paper 1HR • Q21     | $\frac{7}{75}$                                                                  |
| <b>Q50</b> | Edexcel 4MA1 • June 2025 • Pa-                | $\frac{11}{4}$                                                                  |